NAR Genomics and Bioinformatics, 2026, , 1–17

Layered representation diagnostics for ultra-short metagenomic reads

Ruixiang Mei [1] and Jianhua Huang [1]

[1]School of Data Science, The Chinese University of Hong Kong, Shenzhen, Shenzhen 518172, Guangdong, China
*Correspondence: Jianhua Huang, jhuang@cuhk.edu.cn

Abstract

Short-read metagenomic next-generation sequencing (mNGS) pipelines often process trimmed, ambiguous or locally perturbed reads. Useful evidence can be compressed before a classifier or database lookup is applied, so downstream accuracy alone does not reveal which sequence attributes were retained in the representation layer. Here, we present a representation-diagnostics framework for controlled 69–150 bp short-read settings. The framework separates canonical local k-mer composition, global biochemical summaries and coarse positional property pooling, and evaluates their stability, compactness and local-change readability under paired perturbations. CK4P-MSP combines a reverse-complement canonical 4-mer block (K), a global property block (P) and multi-scale property pooling (MSP) in a 222-dimensional block-normalized representation. A seven-group matched ablation revealed metric-specific roles. Property-only summaries had the lowest numerical drift but weakened nearest-clean retrieval; conditional contrasts assigned P primarily to global stability, MSP to grouped local-change readability and K to composition-linked retrieval. CK4P-MSP provided the most balanced block-decomposable profile, with mean paired cosine 0.989, mean drift 0.129 and grouped local-versus-noise macro-F1 of 0.979. Full-position encodings remained higher-dimensional upper bounds for fine positional information, and compressed high-k vector controls were less stable at the same 222-feature budget. These results show that compact biochemical and coarse positional summaries can provide interpretable representation-level audit coordinates alongside, rather than in place of, database-linked exact matching and alignment.

Keywords metagenomics, short reads, representation diagnostics, k-mer features, biochemical sequence encoding

Metagenomic next-generation sequencing can interrogate complex or unexpected microbial mixtures without a fixed target panel, and it has become an important route for infectious-disease investigation when targeted assays are insufficient [1, 2, 3]. Yet the reads entering downstream analysis are often short, trimmed, ambiguous and locally corrupted. Adapter removal, quality trimming and low-biomass contamination can all change the effective sequence evidence available to a pipeline before classification begins [4, 5, 6]. In controlled short-read (69-150 bp) mNGS settings, these physical constraints do not mean that classification is impossible. They can still reduce the identity margin relative to longer reads and make it harder to see which categories of information remain available. A read representation may already have compressed local identity, biochemical regularity or positional information before a downstream benchmark reports success or failure. For this reason, short-read representation is not merely a preprocessing step. It defines which categories of information remain available to any later classifier, alignment routine or database query. In this manuscript, the 69 and 75 bp settings are used only as representative lower-read-length conditions; the underlying clinical data are not publicly available and are not used as experimental input.

Current metagenomic workflows are strongest when they preserve exact or near-exact identity evidence. Canonical k-mer, minimizer, alignment and database-backed systems such as *Kraken*-family tools, *Centrifuge*, *CLARK*, *Kaiju* and related indexing frameworks have established the practical value of explicit sequence matching for taxonomic assignment and downstream biological interpretation [7, 8, 9, 10, 11]. Community benchmarks further show that performance depends on database coverage, sample composition, novelty and evaluation context, not only on the classifier name [12, 13]. These methods should therefore be acknowledged rather than displaced: exact identity evidence remains the backbone for tasks such as species discrimination, allele-sensitive interpretation and resistance-gene calling. Spaced-seed indexing further extends this line by tuning match patterns for sensitivity under mutation, but its goal is still matching accuracy rather than layered diagnostic analysis [14]. The value of additional audit coordinates can become more visible in 69-75 bp lower-read-length settings, where exact identity evidence has less redundancy than at 150 bp. At the same time, these systems are designed to answer a different question from the one asked here. They determine what a read matches, but they do not directly diagnose which kinds of information remain visible in the representation itself when reads are short, noisy or partially degraded.

Vectorized k-mer representations provide the first broad alternative to explicit matching. Here the central operation is to collapse each read or fragment into a composition profile, such as count, frequency, relative-frequency, TF-IDF, or background-adjusted features. PlasFlow and the BMC Bioinformatics 2018 bacteria classifier show that this compact route can be highly competitive when the task is composition-driven rather than order-driven [15, 16]. Resource-saving k-mer classification extends the same logic by showing that low-dimensional k-mer distributions plus classical machine learning can remain competitive even when deep models are not the only option [17]. However, these vectorized k-mer methods rely solely on composition statistics and do not explicitly layer biochemical stability or positional information alongside identity evidence. That distinction becomes more consequential in 69-75 bp reads, where local composition survives only in compressed form. These methods are not weak baselines; they are a distinct representation family whose success depends on how much local composition survives the short-read regime.

Hashing and sketching form a separate compression family rather than a lower-rank version of SVD. MinHash, locality-sensitive hashing (LSH), histosketch and related strategies map k-mer sets or profiles into fixed-length signatures for rapid similarity search, large-scale database indexing or ultra-lightweight compression [18, 19, 20, 21]. Kssd and the HULK/histosketch line compress representations in service of indexing or neighborhood search, not in service of preserving linear variance structure. That distinction matters here because our vector route is not asking for the cheapest signature possible; it asks whether a sparse composition vector retains signal accessible to a fixed shallow probe once the short-read regime is held constant. Sketching papers therefore belong in the related work as a neighboring compression paradigm, not as a substitute for vectorized k-mer analysis.

Biochemical and signal-based encodings add a second alternative representation family. Chaos-game and genomic-signal formulations, together with numerical mappings such as Voss, tetrahedron, and electron-ion interaction potential (EIIP)-style encodings, expose structure that is not reducible to literal token composition alone [22, 23, 24, 25, 26, 27, 28]. These encodings matter because they keep the question on the representation itself: what survives after a read is translated into a signal, not only what survives after it is matched. In lower-read-length settings, such encodings are especially relevant because they can expose stability patterns when database-linked identity evidence has less redundancy. However, existing biochemical encodings are usually used as standalone features rather than layered alongside local k-mer-composition evidence, and their perturbation-stabilizing value under controlled short-read conditions remains largely unquantified.

Sequence-token deep learning classifiers make the neighboring problem space especially relevant. DeepMicrobes introduced deep learning for metagenomic taxonomic classification, using canonical k-mer tokenization, embedding, BiLSTM and attention for read-level assignment [29]. MetaTransformer extended the same general family with self-attention and alternative tokenization or embedding schemes, including character-level, BPE, k-mer and hash/LSH variants [30]. BERTax showed that a small-k Transformer can support hierarchical taxonomy if the pretraining and output heads are aligned to the task [31]. BarcodeBERT then demonstrated that barcode-scale biodiversity tasks can benefit from domain-specific pretraining and masking choices, but its setting is still a barcode-specific classification problem rather than shotgun short-read mNGS [32]. Large pre-trained DNA language models further extend this representation family as general-purpose encoders [33, 34, 35, 36, 37]. These learned embeddings can be strong, and in some downstream probes they may outperform compact hand-defined representations. The question asked here is different. We do not test whether CK4P-MSP is a stronger predictive encoder than a Transformer. We ask whether a low-dimensional, training-free and block-decomposable representation can make identity, biochemical and coarse positional evidence separately inspectable under controlled short-read perturbation.

Raw-sequence and position-aware models complete the picture. DeepVirFinder, Virtifier, and DETIRE demonstrate that CNN or hybrid architectures can extract useful motif- and composition-level signal from viral fragments, but their target is viral detection rather than multi-class short-read taxonomy [38, 39, 40]. KmerAperture adds an important boundary case by showing that ordinary k-mer set comparisons can lose synteny, whereas matching k-mers back to their original lists can retain positional information about core and accessory differences [41]. Collectively, these advances matured each representation family in isolation, but they still do not provide a unified diagnostic framework that decomposes ultra-short-read representations into local composition, biochemical and positional channels under controlled perturbation.

Against this background, we study short-read representation as a representation-audit problem rather than as a single-model accuracy competition. We ask how local k-mer-composition retention, biochemical perturbation stability, positional readability and compactness trade off across controlled short-read settings. Our compact main method, CK4P-MSP, combines reverse-complement canonical 4-mer composition with biochemical summaries and multi-scale property pooling to produce a low-dimensional, block-decomposable audit coordinate system. Full-position matrices serve as positional upper bounds that reveal what fine-grained positional resolution can add, whereas the canonical spaced-property (CSP) control is retained as a mechanistic boundary comparator for testing whether matching-oriented spaced-seed priors transfer to dense read-level representations. The study incorporates all seven non-empty K/P/MSP combinations, prespecified conditional contrasts, same-dimension reduction controls, paired non-parametric tests, block-weight audits, k-nearest-neighbour mutual-information robustness checks, dimension-matched high-k compressed k-mer baselines and local mutation-fraction sweeps. The aim is to identify the metric-specific contribution and operating boundary of each block under a compact, interpretable feature budget, without assuming that the blocks are statistically independent or universally necessary.

Materials and Methods

Study design

We designed the study as a representation-diagnostic analysis of controlled short-read perturbation settings. Here, representation diagnostics denotes a prespecified set of representation-level measurements rather than a clinical decision rule: paired stability, nearest-clean retrieval, grouped local-change readout, feature dimension and blockwise decomposition. The central question was whether biochemical and coarse positional summaries add auditable evidence to a canonical local k-mer-composition backbone. CK4 and CK5 were reverse-complement canonical k-mer-composition baselines; P denoted global biochemical-property summaries; MSP denoted multi-scale positional property pooling. The seven non-empty K/P/MSP combinations were evaluated, full-position matrices were treated as positional upper bounds, and the canonical spaced-property (CSP) control served as a spaced-seed mechanism comparator.

This design makes two questions testable. First, whether a compact mixed representation remains close to its clean counterpart under nuisance perturbation while retaining representation-level readability. Second, whether the observed behavior can be assigned to metric-specific K, P or MSP contributions rather than to dimensionality, normalization or dilution in a mixed L2 space. We therefore combined a seven-group ablation with a 2×2 spatial-localization-by-substitution-chemistry audit, same-dimension principal-component analysis (PCA) and singular-value decomposition (SVD) controls, dimension-matched high-k compressed k-mer baselines, block-weight and property-scaling sensitivity, paired non-parametric tests, k-nearest-neighbour mutual-information robustness checks, P/MSP relation analysis, feature-extraction runtime and error-aware ART simulator strata. Shallow readout probes tested whether signal remained accessible to a fixed low-capacity model; they were not treated as production classifier performance. Throughout, mixed-space L2 is interpreted as standardized representation drift after block construction and normalization, not as a natural biophysical distance between commensurate physical units.

The 69 and 75 bp settings fall within a commonly reported 50–75 bp mNGS read-length range [42, p. 99] and correspond to aggregate pre- and post-quality-control mean-length conditions observed in a restricted local sequencing context. Only these aggregate length conditions informed the design; no patient-level reads, labels, identifiers or sequence content were analysed. The two lengths were used as lower-read-length stress conditions, not as a claim that short reads are unusable. The remaining lengths were selected for sensitivity analysis: 100, 125 and 150 bp test whether the trends persist as reads become less extreme, and 300 bp is used only as an extended-read reference in selected boundary analyses.

Representation families

The representation families were defined to separate local k-mer composition, global biochemical summaries and coarse positional information. CK4 and CK5 count reverse-complement canonical contiguous k-mers. P contains the read-level mean and population standard deviation of hydrogen-bond, GC, purine and electron-ion interaction potential (EIIP) maps, together with N fraction, length divided by 200 and Shannon entropy over A/C/G/T/N divided by $\log_2 5$. MSP mean-pools five per-base maps (hydrogen bond, GC, purine, EIIP and N indicator) over relative-position binsets with 2, 3, 4 and 6 bins. For a read of length L and a scale with b bins, the jth bin spans indices $[\lfloor jL/b \rfloor, \lfloor (j+1)L/b \rfloor)$; any empty bin is represented by zeros. Ambiguous bases contribute zero to the four biochemical maps and one to the N-indicator map. The complete coordinate definitions are reported in Supplementary Table S6. MSP does not create a larger sequence vocabulary or reconstruct full position; it adds a coarse positional property summary to an otherwise order-poor k-mer-frequency vector. The default 2+3+4+6 binset was prespecified as a coarse-to-fine relative-position summary whose finest scale remains above single-position resolution across the 69–150 bp range. The seven-group ablation comprised K (136 dimensions), P (11), MSP (75), K+P (147), K+MSP (211), P+MSP (86) and K+P+MSP (222). Full-position composition or property matrices retained per-position information and were used only to estimate an upper bound on positional readability. CSP combines spaced-seed counts with property summaries and was used to test whether matching-oriented spaced-seed priors transfer into dense representation diagnostics.

For mixed representations, each block was computed separately and internally L2-normalized before weighted assembly. For read s_i, let $\mathbf{K}_i$, $\mathbf{P}_i$ and $\mathbf{M}_i$ denote the reverse-complement canonical 4-mer composition block, the global biochemical-property block, and the multi-scale positional-property block, respectively. The internally normalized blocks were defined as

$$\hat{\mathbf{K}}_i = \frac{\mathbf{K}_i}{\|\mathbf{K}_i\|_2}, \quad \hat{\mathbf{P}}_i = \frac{\mathbf{P}_i}{\|\mathbf{P}_i\|_2}, \quad \hat{\mathbf{M}}_i = \frac{\mathbf{M}_i}{\|\mathbf{M}_i\|_2},$$

with zero-norm safeguards applied before normalization. The assembled compact representation was then

$$\mathbf{x}_i = \frac{1}{\sqrt{\alpha^2 + \beta^2 + \gamma^2}} \left[\alpha\hat{\mathbf{K}}_i \,;\, \beta\hat{\mathbf{P}}_i \,;\, \gamma\hat{\mathbf{M}}_i\right],$$

where α, β and γ are block weights and $[\cdot\,;\,\cdot]$ denotes concatenation. Unless otherwise stated, all three weights were set to 1. The block-weight audit compared equal weights with K-dominant $(2, 1, 1)$ and property-dominant $(1, 2, 2)$ settings; the separate MSP sensitivity audit evaluated $\gamma \in \{0, 0.25, 0.5, 1, 2\}$. Because the block norms are fixed before concatenation, the denominator is a fixed scalar for a given weight setting rather than a read-specific global norm. The reported global drift metric was

$$d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2\,,$$

which should be interpreted as standardized representation drift in the assembled feature space, not as a natural physical distance between commensurate biological units. Block normalization fixes the declared contribution of each block but does not make the underlying coordinates physically commensurate. This definition also gives the block-normalized distance identity

$$\begin{aligned} d^2_{\mathrm{mix}}(i, j) = \frac{1}{\alpha^2 + \beta^2 + \gamma^2} \Big[&\alpha^2 \left\|\hat{\mathbf{K}}_i - \hat{\mathbf{K}}_j\right\|_2^2 \\ &+ \beta^2 \left\|\hat{\mathbf{P}}_i - \hat{\mathbf{P}}_j\right\|_2^2 \\ &+ \gamma^2 \left\|\hat{\mathbf{M}}_i - \hat{\mathbf{M}}_j\right\|_2^2 \Big]. \end{aligned}$$

For comparison with the composition-only drift $d_K(i, j) = \|\hat{\mathbf{K}}_i - \hat{\mathbf{K}}_j\|_2$, the mixed squared drift is lower than d_K^2 when

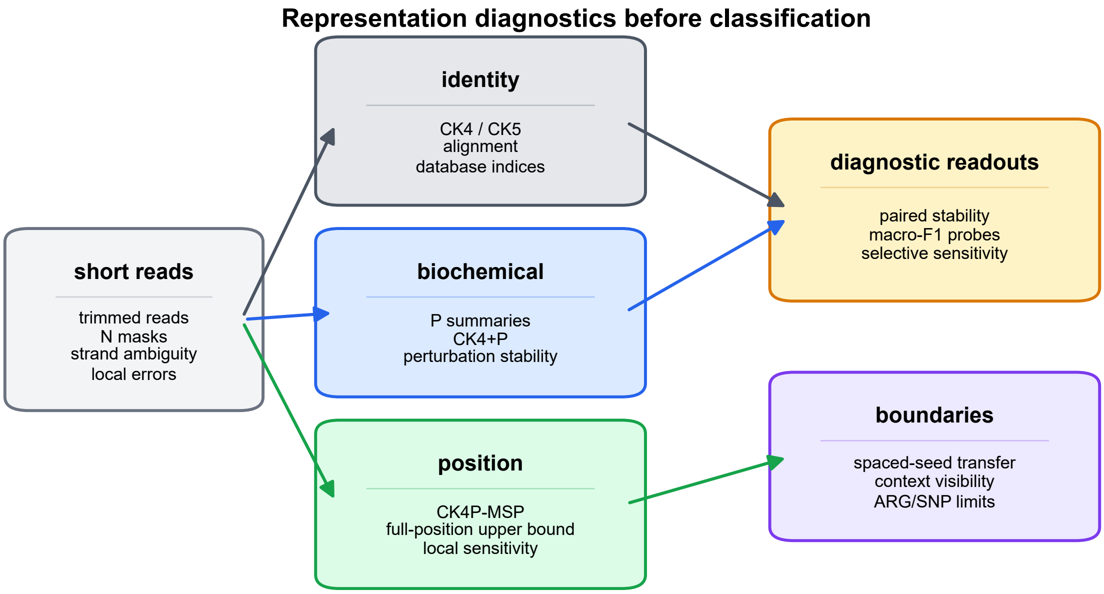

Figure 1 Representation-diagnostic framework and read-length regime. Clean and perturbed reads are encoded through canonical local k-mer composition (K), global biochemical summaries (P) and multi-scale positional property pooling (MSP). The audit reports paired stability, nearest-clean retrieval, grouped local-change readout and feature dimension across 69–150 bp conditions; full-position encodings and the canonical spaced-property (CSP) control provide boundary comparisons.

Alt text: Workflow diagram showing short reads separated into local-composition, biochemical and positional representation channels before diagnostic readouts and boundary checks.

$$\beta^2\left(d_P^2 - d_K^2\right) + \gamma^2\left(d_M^2 - d_K^2\right) < 0,$$

where d_P and d_M are the corresponding blockwise drifts. Thus, lower property and MSP drift than K drift is a sufficient, but not necessary, condition for lower mixed drift. The diagnostic protocol takes matched clean/perturbed reads, constructs each block without fitting a predictive encoder, reports paired stability and nearest-clean retrieval, tests structured local change with grouped delta-readout, and decomposes the result through the seven-group ablation and prespecified conditional contrasts. This protocol yields a representation profile rather than an automated acceptance threshold. The design therefore does not assume that concatenation is intrinsically useful; it requires the combination to survive dimensionality, weight, counterfactual and block-contribution audits.

Data layers and perturbation design

The experiments were organized into eight data layers, with the scale of each layer reported in Table 2. The WGS perturbation layer contained 25,200 rows from 3,600 clean templates across six genera, 21 species and six read lengths; all species and NCBI assembly accessions are listed in Supplementary Table S7. Clean windows were sampled from the public assemblies with seed 303 and expanded into clean, 1% substitution, 3% N masking, 5-bp end trimming, combined 1% substitution plus 3% N masking, 1% short-indel and contiguous 6-bp mismatch conditions. The position-property ablation layer expanded this to 36,000 rows over the same template set and length grid. ART Illumina-like simulation contributed 75,592 clean-versus-simulated rows from 37,796 source templates across 69, 75, 100, 125 and 150 bp [43]. Single-end reads were generated with the ART Illumina executable's default error profile, 0.005-fold coverage, SAM output and a base seed of 606 offset by read length; "paired" in this manuscript refers to each simulated read and its aligned clean template, not paired-end sequencing. The current-contract Critical Assessment of Metagenome Interpretation (CAMI) TOY low-complexity probe used 24,000 labelled 100-bp source reads and expanded them to 216,000 rows at 69, 75 and 100 bp under clean, 3% N masking and 1% substitution conditions [12]. The CAMI II marine lightweight probe used 4,000 anonymous source reads and 48,000 length/condition rows at 69, 75 and 100 bp [13]. The local mutation layer used 250 triplets per analysis cell, four local modes and three read lengths. WGS-derived paired perturbations measured clean-versus-perturbed stability; ART added simulator-derived errors; CAMI resources supplied bounded external probes; controlled position and order tasks tested positional readability; and boundary probes tested spaced-seed transfer, context visibility and ARG/SNP limits. Experiment-specific seeds and complete argument sets are retained in the machine-readable run manifests.

For CAMI_TOY_low, read identifiers were joined to the available taxon mapping and normalized to a common taxon-identifier label field. Thirty labels met the inclusion rule in the sampled source table. Reads without a mapped label were

Table 1. Representation families and diagnostic roles.

family	information channel	main role	main-text interpretation
CK4 / CK5	canonical local k-mer composition	composition backbone and comparator	compact token-frequency evidence, not database-linked read identity
P	global biochemical summaries	perturbation-stable channel	tests global biochemical response without local composition
MSP	positionalized biochemical summaries	coarse positional channel	tests local-change readability without K or global P
CK4+P	composition plus global biochemical signal	global property baseline	tests whether biochemical summaries add to CK4
CK4+MSP	composition plus positionalized biochemical signal	positional property ablation	tests whether MSP returns coarse layout information without global P
P+MSP	global plus positionalized biochemical summaries	property-only combination	tests property stability and retrieval without K
CK4P-MSP	composition plus multi-scale biochemical layout	compact main representation	balances stability, readability and composition-linked retrieval
full-position matrix	per-position token or property signal	upper-bound diagnostic	tests fine positional information at higher dimensional cost
CSP	spaced-seed prior plus biochemical summary	spaced-seed transfer comparator	delimits where matching-oriented spaced seeds transfer to dense features

excluded from the 30-label probe; the binary target-versus-background probe instead used the prespecified target flag, retaining both classes. Per-label sampling was capped before length and perturbation expansion to limit dominance by abundant labels. All four representations in this probe were generated through the current public contract. The 30-label result is reported as a difficult external readability boundary, not as evidence of fine taxonomic classification performance.

We also added a lightweight CAMI II marine subset as an external short-read stability probe. We streamed a prefix of CAMI II marine short-read sample 0 and parsed 4,000 anonymous 150 bp reads from the embedded `anonymous_reads.fq.gz` member. These reads were truncated to 69, 75 and 100 bp and expanded into clean, `N_3pct`, `substitution_1pct` and `substitution_1pct_N_3pct` conditions, yielding 48,000 length/condition rows. CAMI II provides benchmark truth resources, but the anonymous FASTQ headers did not provide read-level taxonomic labels in the streamed-read file used here, and the separate OTU-specific truth bundles were not reconstructed for this lightweight analysis. This CAMI II probe therefore reports paired perturbation-stability metrics on an external metagenomic sequence source rather than CAMI II taxonomic validation.

This layered simulation design was used because each component isolates a different failure mode. The controlled WGS grid makes perturbation type, read length and representation family directly comparable. ART adds a recognized Illumina-like simulator layer, but it is retained as a consistency check rather than treated as ground-truth clinical sequencing physics. The quality-stratified ART audit adds a more explicit error-aware view of the same simulator-derived layer. It supports statements about robustness across quality strata, but it does not by itself establish validation on clinical sequencing data.

Metrics and statistical analysis

Perturbation stability was quantified by paired cosine similarity, paired L2 drift and nearest-clean retrieval. Retrieval searched at most 500 clean candidates within each matched analysis cell; its random-candidate chance level was therefore approximately $1/n_{\mathrm{pool}}$, and saturation at 1.0 limits its ability to rank strong K-containing representations. Representation readability was quantified by macro-F1 and accuracy from shallow logistic-regression or nearest-centroid probes. Logistic probes used standardized inputs, scikit-learn's default L2 penalty and $C = 1$, a fixed random seed, and a 70:30 stratified split for global readouts; local-change probes used fivefold stratified grouped cross-validation. The maximum iteration count was 5,000 for manuscript readouts and 4,000 in the dimensional-reduction audit. These readout metrics were accessibility probes for low-dimensional features, not production classifier performance. Compactness was measured by feature dimension. Local mutation analysis used selective-sensitivity ratios and grouped delta-readout; all derivatives of one template were kept in the same fold through template identifiers. For the seven-group audit, three contrasts were prespecified: K+P+MSP versus P+MSP for the conditional K contribution, versus K+MSP for P, and versus K+P for MSP. Mean matched-cell differences were summarized by 10,000 paired bootstrap resamples. Two-sided paired Wilcoxon signed-rank tests were adjusted across the 12 metric-by-contrast rows by the Benjamini–Hochberg procedure [44, 45]. The analysis unit for these tests was the matched length-by-perturbation or length-by-local-mode cell, not an independent biological cohort. Same-dimension PCA/SVD controls used randomized PCA or TruncatedSVD fitted on the training partition at 147- and 222-feature targets with the experiment seed; block-weight audits tested reasonable reweightings of the composition and property blocks.

The mixed-feature question was also tested through empirical mutual-information (MI) audits. For a perturbation label Y and representation-derived distance features X, an equal-frequency discretized screen was retained as a supplementary descriptive check. The primary analysis used a Ross/Kraskov–Stögbauer–Grassberger (KSG)-style nearest-neighbour estimator with $k = 5$ on low-dimensional continuous summaries and a discrete local-versus-nuisance label [46, 47]. It evaluated d_K, $d_K + d_P$, $d_K + d_M$ and $d_K + d_P + d_M$ across 12 local-mutation cells, with 100 shared label permutations and 100 stratified subsamples per cell. Signed joint-minus-K increments were summarized

Table 2. Data layers, diagnostic questions, metrics and claim boundaries.

data layer	scale	diagnostic question	primary metrics	claim boundary
WGS perturbation pairs	25,200 rows; 3,600 clean templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	Which representations keep clean and perturbed reads close?	paired cosine, L2 drift, retrieval	controlled representation stability, not clinical validation
Position-property ablation	36,000 rows; 3,600 templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	What do P and MSP add to the CK4 composition backbone?	paired stability, block drift, delta-readout, MI proxies	decomposable representation audit, not a natural physical metric
ART Illumina-like simulation	75,592 paired rows; 37,796 templates; 6 genera/21 species; lengths 69,75,100,125,150 bp	Do stability trends persist under simulator-derived read errors?	paired cosine, L2 drift, retrieval	platform-like consistency check
CAMI_TOY_low probes	24,000 labelled source reads; 216,000 expanded rows; 30 taxon identifiers; lengths 69,75,100 bp	Is coarse or 30-label information readable in an external metagenomic source?	target/background and 30-label macro-F1	current-contract external readout; not production taxonomic classification
CAMI II marine anonymous-read probe	48,000 rows from 4,000 anonymous source reads; lengths 69,75,100 bp	Do stability trends persist in a more complex external metagenomic read source?	paired cosine, L2 drift, retrieval	paired stability without reconstructed read-level taxonomic labels
controlled position tasks	controlled synthetic tasks over the short-read length grid	What does fine positional resolution add?	macro-F1, feature dimension	upper-bound diagnostic
local mutation sensitivity	250 triplets per analysis cell; 12 representations; 4 local modes; lengths 69,100,150 bp	Can local biochemical change be detected beyond equal-count noise?	sensitivity ratio, delta-readout macro-F1	selective sensitivity, not functional effect prediction
boundary probes	spaced-seed, context-visibility and ARG/SNP boundary screens	Where do spaced-seed, context and ARG/SNP claims stop?	seed stability, visibility threshold, macro-F1	mechanism boundary and failure modes

by a 10,000-resample paired-cell bootstrap and a paired Wilcoxon test; the aggregate permutation P value was calculated from the mean signed increment under the shared null permutations. These estimates are estimator-dependent empirical summaries rather than absolute physical information quantities. The seven-group contribution audit was paired with a relation audit based on rank, canonical-correlation analysis (CCA) and row-wise P/MSP alignment. Together, these analyses test conditional empirical roles; they do not imply that P and MSP measure unrelated physical quantities.

The mechanism-aligned local-change audit crossed spatial organization (contiguous versus dispersed substitutions) with substitution chemistry (property-directed versus uniformly random alternatives). It used the same seven K/P/MSP combinations and template-grouped splits. Two readouts were prespecified: spatial localization (contiguous versus dispersed) and substitution chemistry (property-directed versus random). A separate scaling audit recomputed P and MSP under the declared mappings, fixed per-property $[0, 1]$ maps and train-fit coordinate z-scores. Leave-one-property-family-out variants tested whether one P family alone determined the stability result. These analyses distinguish the intended block roles from a universal claim that every block improves every metric.

We added a dimension-matched high-k compression audit to avoid comparing CK4P-MSP only with short contiguous k-mer baselines. Canonical $k = 15$ signals were compressed to the 222-feature CK4P-MSP budget using hashing-trick counts and sparse random projection. These vector controls were evaluated on the same 69, 75, 100 and 150 bp perturbation grid using paired cosine similarity, L2 drift, nearest-clean retrieval and shallow readout. MinHash was assessed separately in its native estimated-versus-exact Jaccard geometry and was not treated as an L2 vector or linear-readout feature. This audit tests whether compact stability persists against high-specificity k-mer information under a comparable vector budget.

Finally, a feature-extraction runtime audit compared CK4P-MSP with CK4, CK4+P and the compact high-k controls. Runtime was treated as an engineering boundary, not as a primary performance claim.

Reproducibility

The public implementation and contract-v2 result namespace are maintained in the release repository. The manuscript-facing API is `methods/ck4p_msp.py`; the contribution, high-k, KSG, factorial, property-scaling, MSP-sensitivity and redundancy/runtime audits are executed from `experiments/audits/` and mapped to their result tables in `docs/contract_v2_evidence_map.md`. Historical scripts and outputs remain available for provenance but are not aliases for the public block-normalized CK4P-MSP method. Analyses used Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, SciPy 1.16.3, scikit-learn 1.7.2 and Matplotlib 3.10.6 on Windows 11 (build 26200), an Intel Core Ultra 5 125H processor (14 cores/18 threads) and 31.6 GiB RAM. Runtime measurements were single-process wall-clock timings after one untimed warm-up and five timed repetitions. Peak Python-managed allocation was measured in a separate pass with `tracemalloc`; this excludes memory retained solely by native-library allocators. Both are machine-specific engineering measurements rather than hardware-independent complexity estimates.

Results

Empirical mixed-feature information audit under local perturbation

We first tested whether the added property channels showed perturbation-associated signal rather than only changing the scale of a mixed feature space. The contract-v2 KSG-style audit was applied to low-dimensional blockwise distance summaries for local change versus matched nuisance perturbation. Across 12 local-mutation cells, the mean estimate was 0.889 bits for $d_K + d_P + d_M$ and 0.752 bits for d_K alone. The mean signed empirical separability increment was 0.138 bits (paired-cell bootstrap 95% CI 0.073–0.207; paired Wilcoxon $P = 4.88 \times 10^{-4}$; aggregate label-permutation $P = 0.0099$). The increment was strongest in the 69- and 100-bp cells and attenuated at 150 bp, so it should not be read as length-invariant. Under the tested perturbation grid and estimator, P/MSP distance summaries therefore improved local-versus-noise separability beyond the CK4 distance summary.

This audit complements the ablation results: the property blocks contribute empirically detectable perturbation signal under the tested local-change regime, while the block-normalized metric remains standardized representation drift rather than a physical distance with commensurate units.

Estimator sensitivity supports this result (Supplementary Figure S2 and Supplementary Table S3). A separate same-dimension counterfactual tested whether the result followed from adding coordinates with similar marginal scales. Across 20 WGS stability cells, sequence-linked CK4P-MSP had mean drift 0.135, compared with 0.160 after jointly permuting the P/MSP rows and 0.629 for marginally matched Gaussian controls. Across 12 grouped local-change cells, its logistic macro-F1 was 0.979, compared with 0.879 and 0.886 for the permuted and Gaussian controls, respectively. The controls therefore preserved dimensional expansion without preserving the observed sequence-to-property mapping (Supplementary Figure S5).

Global and positional property layers have related but distinct roles

We then evaluated all seven non-empty K/P/MSP combinations over 69, 75, 100 and 150 bp reads. Pure property summaries had the lowest numerical drift (P, 0.025; MSP, 0.027; P+MSP, 0.027), but they did not preserve the same nearest-clean behavior as K-containing representations: retrieval was 0.295 for P and 0.942 for P+MSP, compared with 1.000 for CK4P-MSP. The composition block therefore traded some numerical stability for composition-linked retrieval. Relative to P+MSP, adding K increased retrieval by 0.058 (95% bootstrap CI 0.036–0.081; BH-adjusted $q = 7.3 \times 10^{-4}$) while increasing, rather than decreasing, drift.

Conditional effects differed by audit axis. Adding P to CK4+MSP reduced mean drift from 0.156 to 0.129, a matched-cell improvement of 0.027 (95% CI 0.023–0.031; $q = 2.4 \times 10^{-7}$), but did not materially change grouped delta-readout (0.9780 versus 0.9786; $q = 0.525$). Adding MSP to CK4+P produced a similar drift reduction of 0.027 (95% CI 0.023–0.032; $q = 2.4 \times 10^{-7}$) and increased grouped local-change macro-F1 from 0.904 to 0.979 (mean difference 0.074, 95% CI 0.035–0.117; $q = 7.3 \times 10^{-4}$). Adding K to P+MSP increased macro-F1 by only 0.008 and the interval included zero. These results assign P primarily to global stability, MSP to local-change readability and K to composition-linked retrieval under the tested grid; they do not establish statistical independence or universal necessity.

The redundancy audit also cautions against overinterpreting P and MSP as separable sources. Across the tested lengths, the first canonical correlation between P and MSP was near one and their first principal components were strongly aligned, as expected because both blocks are derived from the same biochemical property family. However, MSP had higher numerical rank than P, and row-wise P/MSP alignment after compression was incomplete. We therefore interpret the blocks as related layers at different scales, not as interchangeable evidence sources or as measurements of unrelated physical quantities.

The 2×2 mechanism audit made that distinction more specific. Conditional on CK4+P, adding MSP increased spatial-localization macro-F1 by 0.065 (95% CI 0.034–0.098; BH-adjusted $q = 9.77 \times 10^{-4}$) and substitution-chemistry macro-F1 by 0.092 (95% CI 0.059–0.130; $q = 9.77 \times 10^{-4}$). Conditional on CK4+MSP, adding P did not improve spatial localization (difference < 0.001; $q = 0.520$) but increased chemistry readout by 0.030 (95% CI 0.013–0.048; $q = 0.0315$). Thus, MSP contributed coarse localization and chemistry-readable structure, whereas P contributed a smaller global chemistry increment without an independent spatial increment. The blocks were empirically complementary on selected axes, not statistically independent or individually necessary for every task (Supplementary Figure S11 and Supplementary Table S4).

The property-scaling audit further bounded the geometric interpretation. CK4P-MSP drift was 0.128 under the declared maps and 0.132 under fixed per-property $[0, 1]$ maps, while train-fit coordinate z-scoring increased drift to 0.390. Leave-one-property-family-out P variants had a narrow drift range of 0.155–0.156. The stability pattern is therefore insensitive to reasonable fixed remapping and is not determined by one P coordinate family, but it is not invariant to arbitrary variance amplification. P and MSP should accordingly be described as encoding-value-weighted summaries (Supplementary Figure S11 and Supplementary Table S5).

The P/MSP relation and runtime audit is shown in Supplementary Figures S8 and S9; the mechanism and scaling audits are shown in Supplementary Figure S11.

The corrected block-specific runtime audit required approximately 6.23 s per 10,000 reads for CK4P-MSP, compared with 1.36 s for CK4, 3.36 s for CK4+P and 2.98 s for the 222-dimensional hashed $k = 15$ control. Peak Python-managed allocation was 2.80 MiB for CK4P-MSP, 2.45 MiB for CK4 and 3.55 MiB for the hashed control. CK4P-MSP is therefore compact in feature dimension and requires no training, but its explicit property extraction is not the fastest tested encoder.

MSP short-bin reliability remains usable at 69-75 bp

We next tested whether MSP becomes noise-dominated when ultra-short reads are divided into finer relative-position bins. The short-bin audit did not show an immediate loss of reliability at 69 or 75 bp. At the pre-specified $\gamma = 1$ setting, the full 2+3+4+6 binset had paired cosine of approximately 0.985 and L2 drift of 0.153 at 69 bp, and paired cosine of approximately 0.987 and L2 drift of 0.140 at 75 bp. Retrieval remained effectively saturated.

The grouped local delta-readout audit did not show an immediate short-bin failure. At 69 bp and $\gamma = 1$, mean macro-F1 increased from 0.876 with 2 bins to 0.907 with 2+3 bins, 0.951 with 2+3+4 bins and 0.959 with 2+3+4+6 bins. The same

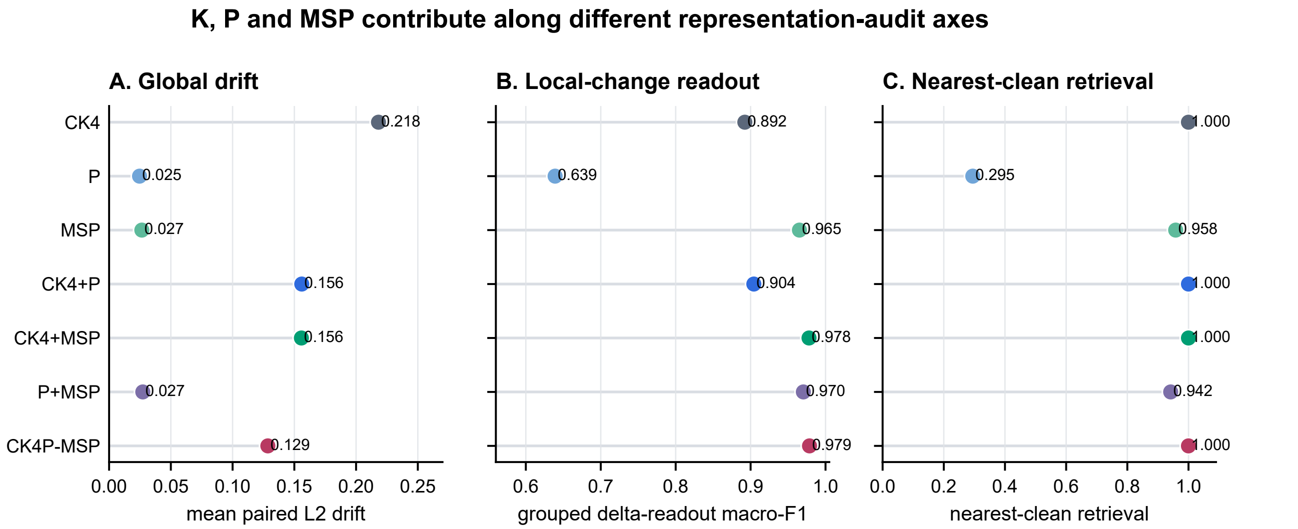

Figure 2 Seven-group K/P/MSP ablation across 24 matched length-by-perturbation cells and 12 grouped local-mutation cells. (A) Mean paired L2 drift is lowest for property-only summaries; CK4P-MSP is more stable than CK4 while retaining the K block. (B) MSP-containing representations preserve the strongest grouped local-versus-nuisance delta-readout under template-grouped fivefold cross-validation. (C) K-containing representations retain near-perfect nearest-clean retrieval from pools of at most 500 candidates, whereas P-only and P+MSP lose composition-linked retrieval. Symbols show means across cells; exact conditional contrasts, 95% paired-cell bootstrap intervals, two-sided Wilcoxon tests and Benjamini–Hochberg-adjusted q values are reported in Supplementary Table S2.

Alt text: Three aligned dot plots compare all seven K/P/MSP block combinations by global paired L2 drift, grouped local-change macro-F1 and nearest-clean retrieval. Property-only summaries have the lowest drift, MSP-containing summaries have the strongest local-change readout, and K-containing summaries have the strongest retrieval.

monotonic pattern persisted at 100 and 150 bp. Increasing γ to 2 further reduced global drift, so the default $\gamma = 1$ should be understood as a fixed balanced setting rather than an empirically optimal stability weight. The combined interpretation is restrained but favorable: static MSP did not become noise-dominated in the tested ultra-short-read regime, yet it remains a coarse positional summary rather than an optimized variance-aware weighting model.

MSP binset and weight sensitivity are shown in Supplementary Figure S6.

Error-aware ART perturbation audit

We next asked whether the external simulator evidence depended on treating ART as one pooled error source. In the quality-stratified ART audit, representative compact controls remained readable across low-, mid- and high-quality bins. The property-aware compact representation preserved high paired cosine and bounded L2 drift across quality strata, and the spaced-property comparator retained retrieval and readout consistency.

These results strengthen, but do not overstate, the ART evidence. They show that the main pattern is not restricted to an undifferentiated simulator summary. They do not claim that ART quality strata capture the full physical error distribution of a clinical sequencing run.

Quality-stratified ART results are shown in Supplementary Figure S3.

Compact biochemical summaries reduce perturbation drift

The main compact-representation audit tested whether property-aware summaries remained closer to their clean counterparts than composition-only or compressed high-k baselines. Across the WGS-derived perturbation grid, CK4P-MSP retained paired cosine of 0.989 and mean L2 drift of 0.129 at 222 features, compared with 0.969 and 0.220 for CK4 (136 features) and 0.945 and 0.295 for CK5 (512 features). CK4+P retained paired cosine of 0.984 and drift of 0.157 at 147 features. Its conditional comparison with CK4+MSP, rather than an obsolete implementation alias, supports the role of P in global stability.

Dimension-matched high-k vector controls further addressed whether a conventional high-specificity k-mer representation could show the same behavior under the CK4P-MSP feature budget. At approximately 222 features, CK4P-MSP retained paired cosine 0.989 and L2 drift 0.129, compared with 0.821 and 0.524 for hashed $k = 15$ counts and 0.868 and 0.447 for sparse random projection of $k = 15$ counts. Shallow-readout differences were modest and did not favour CK4P-MSP over all high-k controls, so the result supports a compact perturbation-stability advantage over these training-free compressed vectors rather than a predictive-performance claim. MinHash was assessed separately through its native estimated-versus-exact Jaccard agreement (mean estimated Jaccard 0.711, exact Jaccard 0.711, mean absolute error 0.021); it was not used as an L2-vector baseline. Paired Wilcoxon tests over matched analysis cells supported the primary stability contrasts after multiple-testing correction.

Training-fitted reductions defined a separate boundary. CK7 PCA/SVD projections fitted to the clean partition at 147 or 222 target dimensions achieved mean drift of 0.093–0.098,

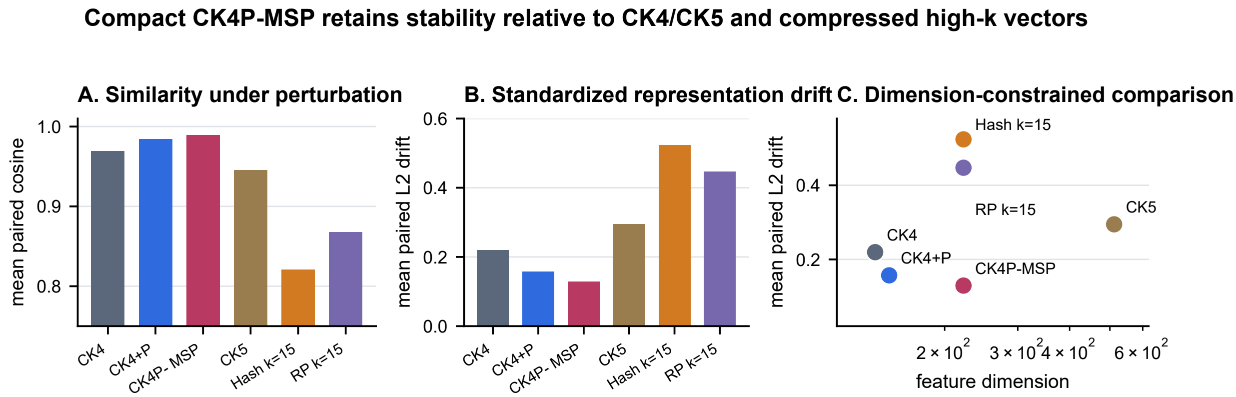

Figure 3 Dimension-matched compact stability audit over 24 length-by-perturbation cells from the controlled WGS grid. Panels compare mean paired cosine similarity, standardized L2 drift and feature dimension across compact composition/property representations and 222-dimensional hashed or sparse-random-projection $k = 15$ vectors. Bars and points show means across matched cells. CK4P-MSP is more stable than CK4, CK5 and both compressed high-k vector controls; lower drift and higher cosine indicate greater paired stability. MinHash is excluded from this L2-vector comparison and evaluated in native Jaccard geometry (Supplementary Figure S7).

Alt text: Bar plots and a scatter plot comparing compact k-mer, property-aware and dimension-matched high-k vector representations by paired similarity, standardized representation drift and feature dimension.

lower than CK4P-MSP, while their logistic readout remained task-limited (macro-F1 0.403–0.416; Supplementary Figure S1). Thus, learned linear projection can optimize numerical stability more strongly than the fixed biochemical construction. CK4P-MSP's distinguishing property is not universal minimum drift; it is a training-free, block-decomposable representation whose K, P and MSP contributions remain directly attributable.

The fitted-reduction boundary and fixed-contract block-weight audit are summarized in Supplementary Figure S1.

Compressed high-k and native MinHash results are provided in Supplementary Figure S7.

CK4P-MSP balances compactness, stability and representation readability

We next evaluated CK4P-MSP as a compact trade-off rather than as a single-metric winner. The seven-group result showed why the complete representation contains all three blocks: property-only variants were numerically stable but weakened nearest-clean retrieval, P improved global stability after conditioning on K+MSP, and MSP improved local-change readout after conditioning on K+P. CK4P-MSP therefore preserves a compact canonical local-composition backbone while exposing global biochemical and coarse positional audit channels.

The resulting trade-off supported the intended role. CK4P-MSP remained compact, reduced perturbation drift relative to CK4 and CK5 composition baselines, retained near-perfect nearest-clean retrieval and preserved accessible local-change signal. Its role is to provide a block-decomposable audit coordinate system between composition-only compact features and high-dimensional full-position matrices, rather than to maximize predictive accuracy against learned embeddings.

External ART and CAMI probes separate stability from task-limited readability

ART and CAMI were used to test whether the representation family showed consistent behavior outside the controlled WGS grid while keeping stability and readout as separate questions. ART supplied simulator-derived paired-error context. Under the current contract, CAMI II marine supplied anonymous-read stability and CAMI_TOY_low supplied coarse and 30-label readability probes. Across nine CAMI II length-by-perturbation cells, mean drift was 0.144 for CK4P-MSP, compared with 0.175 for CK4+P, 0.243 for CK4 and 0.325 for CK5. In the CAMI_TOY_low binary target-versus-background probe, mean logistic macro-F1 was 0.863 for CK4P-MSP, 0.842 for CK4+P, 0.840 for CK4 and 0.857 for CK5. The advantage did not extend to the 30-label probe: mean logistic macro-F1 was low for every compact representation (0.229–0.244), with CK4P-MSP at 0.235. These results support external stability and coarse readability while identifying a fine-label boundary.

The quality-stratified ART audit and the CAMI II marine subset provide different external checks. The former stratifies a simulator-derived error layer by read quality; the latter applies current-contract paired perturbations to anonymous public marine reads without reconstructing read-level taxonomic labels. Marine composition can differ materially from pathogen-rich or low-biomass mNGS sources, including in GC distribution and sequence complexity. The marine result therefore tests whether the stability ordering survives a composition-shifted public source; it is not external taxonomic validation.

Together, ART supports simulator context, CAMI_TOY_low supports task-limited external readability, and CAMI II supports anonymous-read stability. The weak 30-label CAMI result prevents extrapolation to fine taxonomic assignment, while the current-contract coarse and stability results show that the controlled-grid pattern is not confined to one WGS-derived source.

The complete CAMI II marine length-by-perturbation matrix is provided in Supplementary Figure S10.

Table 3. Compact main-method metrics. Paired cosine and L2 drift are means across 24 controlled WGS length-by-perturbation cells. Logistic macro-F1 is averaged across 132 fixed-split WGS readout cells and is a shallow accessibility probe rather than production-classifier performance. Feature count reports the fixed representation dimension.

representation	paired cosine	L2 drift	logistic macro-F1	features
CK4	0.969	0.220	0.363	136
CK4+P	0.984	0.157	0.364	147
CK4P-MSP	0.989	0.129	0.376	222
CK5	0.945	0.295	0.396	512

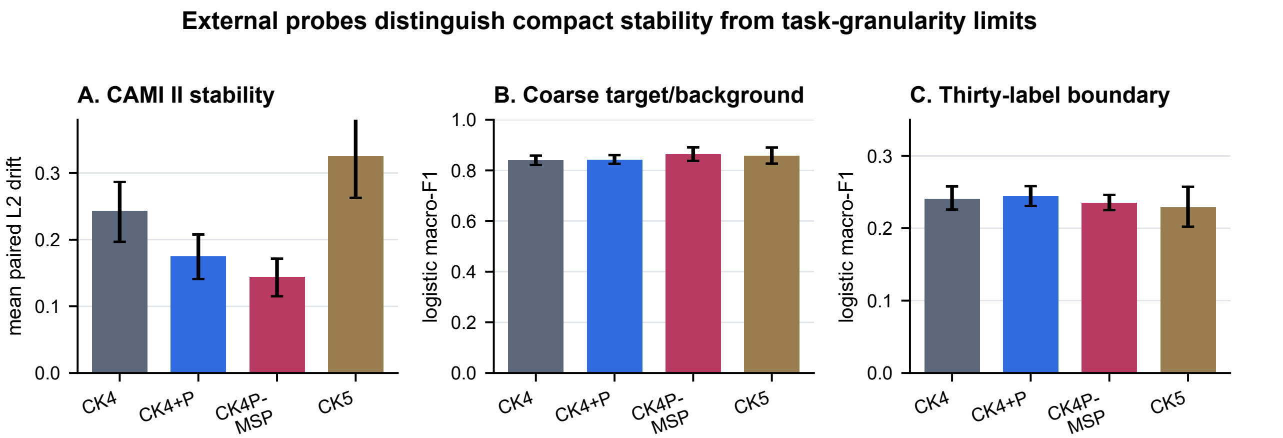

Figure 4 Current-contract external probes separate stability from task-limited readability. (A) Mean standardized L2 drift across nine CAMI II marine length-by-perturbation cells (4,000 anonymous source reads expanded to 48,000 rows); lower is preferable. (B) Mean logistic macro-F1 across nine CAMI_TOY_low length-by-condition cells for binary target-versus-background readout. (C) The corresponding 30-label probe, in which all compact representations remain weak and CK4P-MSP does not lead. Error bars are 95% bootstrap intervals across analysis cells. CAMI II lacks reconstructed read-level labels in this subset, whereas both CAMI_TOY panels use current-contract features from 24,000 labelled source reads expanded to 216,000 rows.

Alt text: Three bar panels comparing current-contract CAMI II anonymous-read drift, CAMI TOY binary target-background readout and the weak CAMI TOY 30-label readout boundary.

Full-position matrices define a positional upper bound rather than a deployment default

Full-position encodings were used to estimate what is gained when fine-grained layout information is retained. In controlled position and order tasks, full-position token or property matrices improved positional readability compared with compact summaries. This was expected, because these matrices preserve per-position structure that compact summaries intentionally compress.

The result is best interpreted as an upper-bound diagnostic. Full-position matrices clarify the value of positional information, but their high dimensionality and task specificity make them unsuitable as the default representation for compact short-read auditing. CK4P-MSP therefore occupies a different role: it does not match the full positional upper bound, but it retains a coarse positional property summary at a much smaller feature cost.

Local mutation analysis separates robustness from selective sensitivity

A stable audit representation should remain responsive to structured changes, but the relevant notion of responsiveness depends on the metric. We therefore tested whether local biochemical change remained distinguishable from matched nuisance perturbation. In a distance-ratio analysis, CK4P-MSP did not exceed CK4: its local-mutation/random-noise L2 ratio was 0.752 compared with 0.750 for CK4. Full-position and property-channel representations were stronger distance-amplification probes. CK4P-MSP's positive result was different: it retained high grouped delta-readout at compact dimensionality, with local-versus-noise macro-F1 of 0.979. The seven-group ablation assigned most of this local-readout increment to MSP, while P contributed primarily to global stability and K to nearest-clean retrieval.

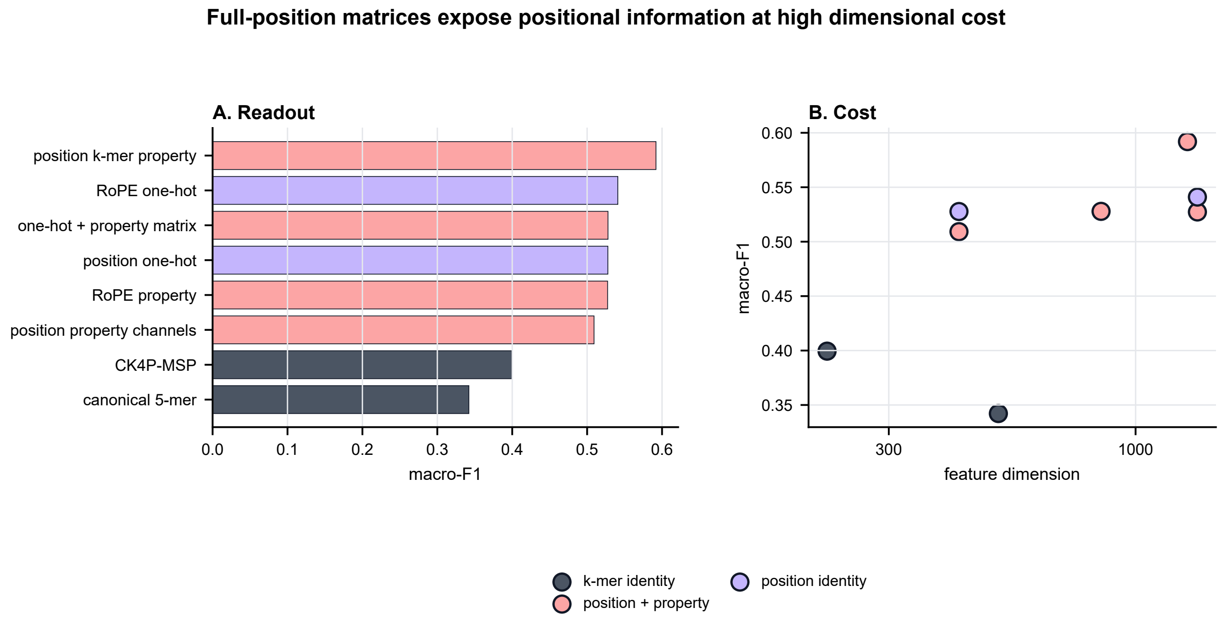

Figure 5 Full-position diagnostic upper bound for positional information. (A) Macro-F1 on controlled positional readout tasks compares compact summaries with representations that retain per-position token or property channels. (B) Readout is plotted against feature dimension to show the cost of preserving near-explicit positional information. CK4P-MSP provides a compact coarse summary and is not the positional-readout upper bound.

Alt text: Bar and scatter plots showing that full-position matrices add positional readout value at substantially higher feature dimension.

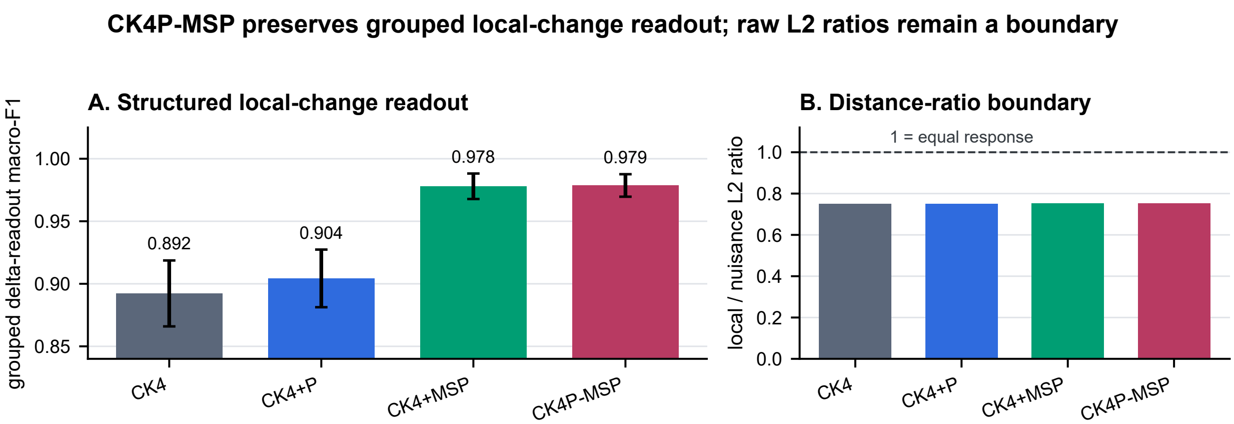

Figure 6 Local-change readout and distance-ratio boundary across 12 length-by-local-mode cells (250 template triplets per cell). (A) Mean local-versus-nuisance macro-F1 under template-grouped fivefold cross-validation shows that MSP accounts for most of the compact local-change readout increment. (B) Mean local-to-nuisance L2 ratio shows that CK4P-MSP does not exceed CK4 on raw distance amplification. Error bars are 95% bootstrap intervals across cells; higher values indicate greater readout or distance selectivity, respectively.

Alt text: Bar plots showing MSP-driven grouped local-change readability and the separate compact-representation distance-ratio boundary.

This result supports a division of labor. Paired cosine and L2 drift quantify nuisance stability, distance ratios identify upper-bound local-change probes, and delta-readout asks whether structured local changes remain accessible to a shallow model. CK4P-MSP should therefore not be described as a distance-amplifying mutation detector. Its value in this analysis is that it remains stable under nuisance perturbation while preserving decomposable information for local-change readout at much lower dimensionality than full-position matrices.

The mutation-fraction sweep is shown in Supplementary Figure S4.

Boundary analyses constrain the interpretation

The boundary probes were included to prevent overextension of the representation claim. Spaced-seed transfer was useful as a mechanism comparator, but matching-oriented seed priors did not automatically become a universal dense-feature advantage. Context visibility depended on whether the relevant relation was present within the read. ARG/SNP interpretation required identity, annotation and allele-level evidence beyond what compact property summaries can provide.

Table 4. Seven-group block ablation. Global drift and nearest-clean retrieval are averaged over 24 matched length-by-perturbation cells. Grouped delta-readout macro-F1 is averaged over 12 length-by-local-mode cells using template-grouped fivefold cross-validation. Lower drift and higher retrieval/readout indicate stronger performance on their respective audit axes.

representation	features	L2 drift	retrieval	grouped macro-F1	observed audit role
CK4	136	0.218	1.000	0.892	local composition and composition-linked retrieval
P	11	0.025	0.295	0.639	global property stability without local-composition retrieval
MSP	75	0.027	0.958	0.965	coarse positional property readout
CK4+P	147	0.156	1.000	0.904	composition plus global property stability
CK4+MSP	211	0.156	1.000	0.978	composition plus local-change readability
P+MSP	86	0.027	0.942	0.970	stable property-only summary with incomplete retrieval
CK4P-MSP	222	0.129	1.000	0.979	balanced block-decomposable trade-off

Table 5. Boundary and mechanism summary.

boundary	main observation	interpretation
spaced-seed transfer	The best tested 4-position dense stability pattern was contiguous 0-1-2-3 across the 69/75 bp sanity grid.	spaced seeds remain useful matching priors, but transfer is regime-specific
context visibility	motif-pair visibility thresholds depended on placement, emerging at 130, 140, 148 or 155 bp.	a representation cannot encode context absent from the read
ARG/SNP readout	stable compact features did not by themselves establish allele, resistance-SNP or functional equivalence.	identity/database evidence remains necessary for final biological calls

These negative and boundary results are part of the manuscript's logic. They keep the claim focused on representation diagnostics and representation-level perturbation auditing, rather than allowing the compact feature space to be misread as a full biological interpretation system.

Discussion

This study treats short-read sequence representation as an inspectable object rather than only as a black-box prelude to classification. Across controlled short-read settings, database-linked exact matching remains distinct from the compact representation itself, while biochemical and coarse positional summaries changed what could be measured under perturbation. CK4P-MSP provided a compact trade-off in this role: it combined a reverse-complement canonical 4-mer composition block with biochemical and multi-scale property-pooling channels that reduced perturbation drift relative to CK4/CK5 and retained local-change readability at compact dimensionality.

The results support a division of labour between representation families. Canonical k-mer counts provide a compact local-composition backbone, while exact matching and alignment retain their distinct database-linked role. Global biochemical summaries provide the strongest global stability channel in the current grid. Multi-scale positional property pooling uses the same biochemical property family to add coarse layout summaries to an otherwise order-poor k-mer-frequency vector and accounts for most of the grouped local-change readout gain. It is not a larger k-mer vocabulary, a learned token mapping or a reconstruction of full positional structure. Full-position matrices, by contrast, define what can be gained when positional information is preserved almost explicitly. They are useful diagnostic upper bounds but are not the preferred compact representation.

The additional analyses sharpen this interpretation. Same-dimension PCA/SVD controls show that numerical stability is not unique to CK4P-MSP: fitted CK7 projections achieved still lower drift, establishing a data-dependent reduction boundary. CK4P-MSP instead offers a fixed, training-free and block-attributable coordinate system. Dimension-matched high-k vector controls show that this fixed representation remained more stable than the tested hashed and random-projection $k = 15$ controls at a comparable feature budget, although shallow readout differences were modest. MinHash was assessed separately in its native Jaccard geometry. Block-weight and fixed-map scaling audits show that the reported trade-off is not tied to one selected weight or reasonable fixed property remapping; the deterioration under train-fit coordinate z-scoring also shows that the geometry is not scale-invariant. Paired Wilcoxon tests show that the principal contrasts are consistent across the tested matched cells. The KSG-style audit indicates an estimator-dependent empirical separability increment beyond CK4 distance in the tested local-mutation regime.

The seven-group contribution audit makes this division of labour more precise. Property-only summaries were numerically stable but did not preserve the same nearest-clean retrieval as K-containing representations. Conditional on K+MSP, P reduced global drift without a detectable increment in the original local-versus-nuisance readout. Conditional on K+P, MSP reduced global drift and produced the largest grouped local-change readout increment. Conditional on P+MSP, K restored nearest-clean retrieval while increasing numerical drift. The factorial audit refined this assignment: MSP improved spatial-localization and chemistry readouts, whereas P added a smaller chemistry increment without a spatial increment. The redundancy audit further showed that P and MSP share a strong global property component but are not interchangeable: MSP has higher numerical rank and incomplete row-wise alignment with P. These results support metric-specific conditional contributions, not statistical independence, universal necessity or a claim that every block improves every endpoint.

The perturbation analyses also clarify the scope of the evidence. Uniform substitutions and local mutation sweeps provide controlled tests of representation behavior. ART adds a simulator-derived Illumina-like layer, and the quality-stratified ART audit shows that the main pattern persists across read-quality strata. CAMI_TOY_low provides current-contract labelled external readout probes: CK4P-MSP showed a small advantage on target-versus-background readout, but no advantage on the weak 30-label task. The CAMI II marine subset provides a more complex anonymous-read external stability probe without read-level taxonomic labels in this lightweight analysis. Its different composition makes the retained stability ordering informative, but does not turn it into pathogen-rich or clinical validation. Together, these data support representation-level stability and bounded readability under the tested conditions.

A useful outcome of the boundary analyses is that they prevent overgeneralization. CSP and spaced-seed controls help interpret which matching-oriented priors transfer into dense read-level representations, but they do not become general substitutes for compact identity-property features. Context and ARG/SNP probes show that compact stability is not the same as allele-level or functional interpretation. Those tasks require database identity, annotation, sufficient context and downstream biological evidence.

Limitations

The first limitation is task scope. The study evaluates representation diagnostics, whereas Kraken 2, Centrifuge, CLARK, Kaiju and related systems remain the appropriate reference frame for database-linked taxonomic assignment. The present work asks what information remains visible before or alongside such assignment systems. Direct integration with production classifiers, including downstream false-hit reduction or classifier-output auditing, remains future work and is not evaluated here.

The second limitation is the perturbation model and external benchmark scale. Controlled substitutions, ambiguous-base perturbations and local mutation sweeps are useful because they isolate representation behavior, but they do not reproduce every physical process in sequencing data. ART and the quality-stratified ART audit partly address this issue by adding simulator-derived and quality-aware evidence. CAMI_TOY_low and CAMI II marine add external metagenomic resources, but with different roles: the former supports lightweight labelled readout, and the latter supports only paired stability because read-level taxonomic labels were not reconstructed for the anonymous reads in this lightweight probe. The resulting conclusions therefore remain bounded to controlled short-read, simulator-supported and external stability/readout settings, not to production-scale community profiling or unobserved clinical error distributions.

The third limitation concerns data provenance. The 69 and 75 bp settings were selected as lower-read-length conditions that fall within a commonly reported post-QC mNGS range, and local restricted sequencing records were not used as study inputs or publicly shared. The experiments therefore use length conditions, not patient-level reads or patient-derived labels. The final submission will include institutionally approved wording for this data-governance boundary.

The fourth limitation is that the MI and conditional-MI analyses are empirical. They support perturbation-associated signal in the tested feature channels, but they are not general evidence that a property-augmented representation always has higher mutual information than a composition-only representation. Likewise, mixed L2 distance is standardized representation drift after block construction and normalization, not a natural metric that equates biochemical properties and k-mer counts as identical physical units. Its magnitude depends on the declared property maps and block weights. Under the tested local-perturbation grid and estimator, property-aware distance summaries showed a positive empirical separability increment relative to composition-only and permutation controls; the weaker 150-bp cells delimit this observation.

The fifth limitation concerns MSP weighting and scale. The short-bin audit did not show an immediate loss of reliability at 69–75 bp, and finer multi-scale pooling increased delta-readout rather than immediately degrading it. Even so, the current MSP channel remains a static coarse summary. It is not an optimized variance-aware estimator, and the manuscript does not claim that its present binset or γ weighting is universally optimal across read lengths, perturbation types or sequencing platforms. The scaling audit further shows that train-fit variance amplification changes the geometry substantially; the reported method is therefore the declared encoding-value-weighted contract, not a scale-free construction.

The sixth limitation is computational scope. CK4P-MSP remained lightweight in absolute terms in the local implementation, but it was slower to extract than CK4 and simple high-k hashed or MinHash summaries. Its value in this manuscript is interpretability and perturbation diagnostics under compact dimensionality rather than engineering speed-up. Large-scale deployment would require optimized implementation and pipeline-level cost-benefit testing.

The seventh limitation is model scope. Large DNA language models and learned metagenomic classifiers provide powerful sequence representations. They may outperform CK4P-MSP in predictive probes, especially when sufficient training data, pretraining and compute are available. CK4P-MSP instead provides a low-dimensional, training-free and block-decomposable coordinate system for auditing local composition, biochemical and coarse positional information under controlled short-read perturbation. Direct comparison with learned embeddings would require a separate benchmark design that reports both predictive accuracy and auditability.

Future work

Future work should test whether CK4P-MSP or related compact summaries improve failure-mode detection when placed alongside database-centred pipelines. Pipeline-facing false-hit reduction or classifier-output auditing would require downstream classifier outputs and explicit operating-point analysis. A useful next step is to test whether composition-linked retrieval, biochemical stability and local-change readability can inform explicit representation-level warning signals.

A second direction is to expand the perturbation model. Platform-specific error-transition matrices, explicit Q-score trajectories and larger simulator grids would make the error-aware analysis more realistic. The current controlled simulations are useful because they isolate representation behavior by read length, perturbation type and feature family; future work should test whether the same conclusions hold under richer platform-specific error processes. When ethically and legally feasible, restricted clinical read collections could then be used for external validation under controlled access, with public release limited to aggregate metadata and reproducible scripts.

A third direction is to make the positional channel more adaptive without changing the paper's current claim. Length-aware or variance-aware MSP weighting could be explored in future work, but such a step should be framed as a new model family rather than as a cosmetic parameter retune. The present results justify static MSP as a usable compact summary; they do not settle the separate optimization problem of how positional bins should be weighted under different noise and length conditions.

A fourth direction is to connect compact diagnostics with learned models. Full-position matrices and DNA foundation-model embeddings are natural candidates for follow-up studies because they may recover richer context and stronger predictive signal than compact summaries. The present work suggests where such models should be audited: not only by final classification accuracy, but also by identity retention, perturbation stability, positional readability and local-change sensitivity.

Conclusion

This study shows that compact short-read representations can be evaluated as interpretable audit objects. In controlled short-read settings, CK4P-MSP combined a canonical local k-mer-composition block with global biochemical and positionalized biochemical channels to reduce perturbation drift relative to CK4/CK5 while preserving grouped local-change readability. The evidence supports a bounded implication: compact biochemical and coarse position-aware summaries can provide block-decomposable coordinates for representation-level perturbation auditing alongside database-linked exact matching and curated reference methods.

Data Availability

All processed data tables, figure source summaries and analysis outputs generated for this study are maintained in a private GitHub review repository at `https://github.com/FairYJmrx/ultrashort-dna-representation-diagnostics/tree/release`. Reviewer access will be provided through the submission system or by adding a journal-supplied account as a read-only collaborator. A public archival DOI will be minted from the frozen release when the authors make the repository public.

Publicly reused resources include simulator- and benchmark-derived materials used for ART, CAMI_TOY_low and CAMI II marine analyses [43, 12, 13]. The CAMI II marine lightweight probe used the public CAMI II marine short-read sample 0 reads archive and its corresponding setup archive; exact resource URLs are listed in the review-access repository and will be retained in the public archived release. Only a streamed prefix was parsed for the lightweight stability probe, and read-level taxonomic labels were not reconstructed from the separate truth resources for this analysis. The generated subset tables, stability summaries and source figure table are included in the review-access repository. Local restricted sequencing records were used only as read-length provenance for the 69 and 75 bp conditions. The underlying clinical sequencing reads were not used as experimental input, are not part of the study data package, and cannot be publicly shared.

Code Availability

The analysis scripts used to generate tables, figures and methodological audits are maintained in the same private review-access repository under an MIT licence. The repository README and release manifest map each manuscript figure, table and supplementary audit to its source script, input tables and generated outputs. Reviewer access will be provided during submission, and a public archived release identifier will be added after public release.

Supplementary Data

Supplementary Data are available at *NAR Genomics and Bioinformatics* Online. The supplementary file includes Supplementary Figures S1–S11 and Supplementary Tables S1–S7. The underlying machine-readable source tables and figure-generation inputs are included in the review-access repository and will be included in the public archived release.

Ethics and Data Governance

No patient-level clinical sequencing reads, patient identifiers or patient-derived labels were analyzed in this study. Institutional wording for this boundary statement must be finalized before submission.

Author Contributions

Ruixiang Mei: Conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing – original draft, and writing – review and editing. Jianhua Huang: Supervision and writing – review and editing. Ruixiang Mei ORCID: `https://orcid.org/0009-0003-2128-0726`.

Funding

Funding information will be finalized before submission.

Acknowledgements

Acknowledgements, if any, will be finalized before submission.

Conflict of Interest

The authors declare no competing interests.

References

1. Michael R. Wilson, Samia N. Naccache, Erika Samayoa, Mark Biagtan, Ali Bashir, Guixia Yu, S. M. Salamat, Sneha Somasekar, Scot Federman, Steve Miller, Robert Sokolic, Elitza Garabedian, Fabio Candotti, Rebecca H. Buckley, Kurt D. Reed, Terry L. Meyer, Christine M. Seroogy, Renee Galloway, Stuart L. Henderson, James E. Gern, Joseph L. DeRisi, and Charles Y. Chiu. Actionable diagnosis

of neuroleptospirosis by next-generation sequencing. *New England Journal of Medicine*, 370:2408–2417, 2014. doi: 10.1056/NEJMoa1401268.

2. Michael R. Wilson, Heather A. Sample, Kaitlyn C. Zorn, Samuel Arevalo, Guixia Yu, John Neuhaus, Scot Federman, Deborah Stryke, Brian Briggs, Charles Langelier, Adam Berger, Victoria Douglas, S. Andrew Josephson, Felicia C. Chow, Blair D. Fulton, Joseph L. DeRisi, Jeffrey M. Gelfand, Samia N. Naccache, Jeffrey M. Bender, and Charles Y. Chiu. Clinical metagenomic sequencing for diagnosis of meningitis and encephalitis. *New England Journal of Medicine*, 380: 2327–2340, 2019. doi: 10.1056/NEJMoa1803396.
3. Charles Y. Chiu and Steven A. Miller. Clinical metagenomics. *Nature Reviews Genetics*, 20:341–355, 2019. doi: 10.1038/s41576-019-0113-7.
4. Marcel Martin. Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal*, 17:10–12, 2011. doi: 10.14806/ej.17.1.200.
5. Anthony M. Bolger, Marc Lohse, and Bjoern Usadel. Trimmomatic: A flexible trimmer for illumina sequence data. *Bioinformatics*, 30:2114–2120, 2014. doi: 10.1093/bioinformatics/btu170.
6. Susannah J. Salter, Michael J. Cox, Elina M. Turek, Szymon T. Calus, William O. Cookson, Miriam F. Moffatt, Paul Turner, Julian Parkhill, Nicholas J. Loman, and Alan W. Walker. Reagent and laboratory contamination can critically impact sequence-based microbiome analyses. *BMC Biology*, 12:87, 2014. doi: 10.1186/s12915-014-0087-z.
7. Derrick E. Wood and Steven L. Salzberg. Kraken: Ultrafast metagenomic sequence classification using exact alignments. *Genome Biology*, 15:R46, 2014. doi: 10.1186/gb-2014-15-3-r46.
8. Derrick E. Wood, Jennifer Lu, and Ben Langmead. Improved metagenomic analysis with kraken 2. *Genome Biology*, 20: 257, 2019. doi: 10.1186/s13059-019-1891-0.
9. Rachid Ounit, Steve Wanamaker, Timothy J. Close, and Stefano Lonardi. Clark: Fast and accurate classification of metagenomic and genomic sequences using discriminative k-mers. *BMC Genomics*, 16:236, 2015. doi: 10.1186/s12864-015-1419-2.
10. Daehwan Kim, Li Song, Florian P. Breitwieser, and Steven L. Salzberg. Centrifuge: Rapid and sensitive classification of metagenomic sequences. *Genome Research*, 26:1721–1729, 2016. doi: 10.1101/gr.210641.116.
11. Peter Menzel, Kim Lee Ng, and Anders Krogh. Fast and sensitive taxonomic classification for metagenomics with kaiju. *Nature Communications*, 7:11257, 2016. doi: 10.1038/ncomms11257.
12. Alexander Sczyrba, Peter Hofmann, Peter Belmann, David Koslicki, Stefan Janssen, Johannes Droege, Ivan Gregor, Stephan Majda, Jessika Fiedler, Eik Dahms, Andreas Bremges, Adrian Fritz, Ruben Garrido-Oter, Tue Sparholt Jorgensen, Nicole Shapiro, Philip D. Blood, Alexey Gurevich, Yang Bai, Dmitrij Turaev, Matthew Z. DeMaere, Rayan Chikhi, Niranjan Nagarajan, Christopher Quince, Fernando Meyer, Monika Balvociute, Lars Hestbjerg Hansen, Soren J. Sorensen, Nicholas Chia, Bertrand Denis, Jeff L. Froula, Zhong Wang, Rob Egan, Dongwan Don Kang, Jeffrey J. Cook, Charles Deltel, Michael Beckstette, Claire Lemaitre, Pierre Peterlongo, Guillaume Rizk, Dominique Lavenier, Yu-Wei Wu, Steven W. Singer, Chirag Jain, Marc Strous, Heiner Klingenberg, Peter Meinicke, Michael D. Barton, Thomas Lingner, Hsin-Hung Lin, Yu-Chieh Liao, Genivaldo Gueiros Z. Silva, Daniel A. Cuevas, Robert A. Edwards, Surya Saha, Vitor C. Piro, Bernhard Y. Renard, Mihai Pop, Hans-Peter Klenk, Markus Goeker, Nikos C. Kyrpides, Tanja Woyke, Julia A. Vorholt, Paul Schulze-Lefert, Edward M. Rubin, Aaron E. Darling, Thomas Rattei, and Alice C. McHardy. Critical assessment of metagenome interpretation - a benchmark of metagenomics software. *Nature Methods*, 14: 1063–1071, 2017. doi: 10.1038/nmeth.4458.
13. Fernando Meyer, Adrian Fritz, Zhi-Luo Deng, David Koslicki, Alexey Gurevich, Gary Robertson, Mohammed Alser, Dmitry Antipov, Francesco Beghini, Denis Bertrand, Jaqueline J. Brito, C. Titus Brown, Jan Buchmann, Aydin Buluc, Bo Chen, Rayan Chikhi, Philip T. L. C. Clausen, Alesia Cristian, Piotr W. Dabrowski, Aaron E. Darling, Rob Egan, Eleazar Eskin, Evangelos Georganas, Eugene Goltsman, Melissa A. Gray, Lars Hestbjerg Hansen, Steven Hofmeyr, Pingqin Huang, Luiz Irber, Hongying Jia, Tue Sparholt Jorgensen, Md. Rezaul Karim, Terje Klemetsen, Axel Kola, Sergey Koren, Jason Kwan, Nathan LaPierre, Claire Lemaitre, Chen Li, Antoine Limasset, Fabio Malcher-Miranda, Serghei Mangul, Vanessa R. Marcelino, Camille Marchet, Pierre Marijon, Dmitry Meleshko, Daniel R. Mende, Alessio Milanese, Niranjan Nagarajan, Jakob Nissen, Sergey Nurk, Leonid Oliker, David Paez-Espino, Pierre Peterlongo, Vitor C. Piro, Jacob S. Porter, Simon Rasmussen, Evan R. Rees, Knut Reinert, Bernhard Y. Renard, Espen M. Robertsen, Gail L. Rosen, Hans-Joachim Ruscheweyh, Varuni Sarwal, Nicola Segata, Enrico Seiler, Lizhen Shi, Fengzhu Sun, Shinichi Sunagawa, Soren J. Sorensen, Torsten Thomas, Chengchen Tong, Mirko Trajkovski, Julien Tremblay, Gherman V. Uritskiy, Riccardo Vicedomini, Zhong Wang, Yuzhen Ye, Pelin Yilmaz, Ronghui You, Georg Zeller, Sen Zhao, Shanfeng Zhu, Shaochun Zhu, Ruben Garrido-Oter, Petra Gastmeier, Stephane Hacquard, Susanne Haussler, Ariane Khaledi, Friederike Maechler, Fantin Mesny, Simona Radutoiu, Paul Schulze-Lefert, Nathiana Smit, Till Strowig, Andreas Bremges, Alexander Sczyrba, and Alice C. McHardy. Critical assessment of metagenome interpretation: the second round of challenges. *Nature Methods*, 19:429–440, 2022. doi: 10.1038/s41592-022-01431-4.
14. Karel Brinda, Michal Sykulski, and Gregory Kucherov. Spaced seeds improve k-mer-based metagenomic classification. *Bioinformatics*, 31:3584–3592, 2015. doi: 10.1093/bioinformatics/btv419.
15. Pawel S. Krawczyk, Leszek Lipinski, and Andrzej Dziembowski. Plasflow: Predicting plasmid sequences in metagenomic data using genome signatures. *Nucleic Acids Research*, 46: e35, 2018. doi: 10.1093/nar/gkx1321.
16. Antonino Fiannaca, Laura La Paglia, Massimo La Rosa, Giosue' Lo Bosco, Antonio Urso, et al. Deep learning models for bacteria taxonomic classification of metagenomic data. *BMC Bioinformatics*, 19(Suppl 7):198, 2018. doi: 10.1186/s12859-018-2182-6.
17. Wolfgang Fuhl, Susanne Zabel, and Kay Nieselt. Resource saving taxonomy classification with k-mer distributions and machine learning, 2023.

18. Brian D. Ondov, Todd J. Treangen, Pall Melsted, Adam B. Mallonee, Nicholas H. Bergman, Sergey Koren, and Adam M. Phillippy. Mash: Fast genome and metagenome distance estimation using minhash. *Genome Biology*, 17: 132, 2016. doi: 10.1186/s13059-016-0997-x.
19. Andrzej Zielezinski, Susana Vinga, Jonas Almeida, and Wojciech M. Karlowski. Alignment-free sequence comparison: Benefits, applications, and tools. *Genome Biology*, 18:186, 2017. doi: 10.1186/s13059-017-1319-7.
20. Hong Yi, Yao Lin, Wei Jin, et al. Sequences dimensionality-reduction by k-mer substring space sampling enables effective resemblance- and containment-analysis for large-scale omics-data. *Genome Biology*, 22:84, 2021. doi: 10.1186/s13059-021-02303-4.
21. Will P. M. Rowe, Anna Paola Carrieri, Cristina Alcon-Giner, Shabhonam Caim, Alexander Shaw, Kevin Sim, J. Simon Kroll, Louise J. Hall, Edward O. Pyzer-Knapp, and Maureen D. Winn. Streaming histogram sketching for rapid microbiome analytics. *Microbiome*, 7:40, 2019. doi: 10.1186/s40168-019-0653-2.
22. H. Joel Jeffrey. Chaos game representation of gene structure. *Nucleic Acids Research*, 18:2163–2170, 1990. doi: 10.1093/nar/18.8.2163.
23. Richard F. Voss. Evolution of long-range fractal correlations and 1/f noise in dna base sequences. *Physical Review Letters*, 68:3805–3808, 1992. doi: 10.1103/PhysRevLett.68.3805.
24. Dimitris Anastassiou. Genomic signal processing. *IEEE Signal Processing Magazine*, 18:8–20, 2001. doi: 10.1109/79.939833.
25. Paul D. Cristea. Conversion of nucleotides sequences into genomic signals. *Journal of Cellular and Molecular Medicine*, 6:279–303, 2002. doi: 10.1111/j.1582-4934.2002.tb00196.x.
26. Gerardo Mendizabal-Ruiz, Israel Román-Godínez, Sulema Torres-Ramos, Ricardo A. Salido-Ruiz, and J. Alejandro Morales. On dna numerical representations for genomic similarity computation. *PLOS ONE*, 12(3):e0173288, 2017. doi: 10.1371/journal.pone.0173288.
27. Gerardo Mendizabal-Ruiz, Israel Román-Godínez, Sulema Torres-Ramos, Ricardo A. Salido-Ruiz, and J. Alejandro Morales. Genomic signal processing for dna sequence clustering. *PeerJ*, 6:e4264, 2018. doi: 10.7717/peerj.4264.
28. Achuthsankar S. Nair and Sivarama Pillai Sreenadhan. A coding measure scheme employing electron-ion interaction pseudopotential. *Bioinformation*, 1:197–202, 2006.
29. Qiaoxing Liang, Paul W. Bible, Youping Liu, Bin Zou, and Li Wei. Deepmicrobes: Taxonomic classification for metagenomics with deep learning. *NAR Genomics and Bioinformatics*, 2:lqaa009, 2020. doi: 10.1093/nargab/lqaa009.
30. Felix Wichmann, Jose R. Zamudio, Roland Eils, and Matthias Schlesner. Metatransformer: Deep metagenomic sequencing read classification using self-attention models. *NAR Genomics and Bioinformatics*, 5:lqad082, 2023. doi: 10.1093/nargab/lqad082.
31. Florian Mock, Fleming Kretschmer, Anton Kriese, Sebastian Böcker, and Manja Marz. Taxonomic classification of dna sequences beyond sequence similarity using deep neural networks. *Proceedings of the National Academy of Sciences*, 119(35):e2122636119, 2022. doi: 10.1073/pnas.2122636119.
32. Pablo Millan Arias, Niousha Sadjadi, Monireh Safari, ZeMing Gong, Austin T. Wang, Joakim Bruslund Haurum, Iuliia Zarubiieva, Dirk Steinke, Lila Kari, Angel X. Chang, Scott C. Lowe, and Graham W. Taylor. Barcodebert: Transformers for biodiversity analyses. *Bioinformatics Advances*, 6(1):vbag054, 2026. doi: 10.1093/bioadv/vbag054.
33. Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana V. Davuluri, and Han Liu. Dnabert-2: Efficient foundation model and benchmark for multi-species genome, 2024.
34. Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam H. Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Hassan Sirelkhatim, Guillaume Richard, Marcin J. Skwark, Karim Beguir, Monica Lopez, and Thomas Pierrot. Nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025. doi: 10.1038/s41592-024-02523-z.
35. Eric Nguyen, Michael Poli, Marjan Faizi, Armin W. Thomas, Camden Birch-Sykes, Michael Wornow, Aman Patel, Charles Rabideau, Stefano Massaroli, Yoshua Bengio, Stefano Ermon, Stephen A. Baccus, and Christopher Re. Hyenadna: Long-range genomic sequence modeling at single nucleotide resolution, 2023.
36. Eric Nguyen, Michael Poli, Matthew G. Durrant, Brian Kang, Dhruva Katrekar, David B. Li, Liam J. Bartie, Armin W. Thomas, Samuel H. King, Garyk Brixi, Jeremy Sullivan, Madelena Y. Ng, Ashley Lewis, Aaron Lou, Stefano Ermon, Stephen A. Baccus, Tina Hernandez-Boussard, Christopher Re, Patrick D. Hsu, and Brian L. Hie. Sequence modeling and design from molecular to genome scale with evo. *Science*, 386:eado9336, 2024. doi: 10.1126/science.ado9336.
37. Veniamin Fishman, Yuri Kuratov, Aleksei Shmelev, Maxim Petrov, Dmitry Penzar, Denis Shepelin, Nikolay Chekanov, Olga Kardymon, and Mikhail Burtsev. Gena-lm: A family of open-source foundational dna language models for long sequences. *Nucleic Acids Research*, 53:gkae1310, 2025. doi: 10.1093/nar/gkae1310.
38. Jie Ren, Kai Song, Chao Deng, Nathan A. Ahlgren, Jed A. Fuhrman, Yi Li, Xiaohui Xie, Ryan Poplin, and Fengzhu Sun. Identifying viruses from metagenomic data using deep learning. *Quantitative Biology*, 8(1):64–77, 2020. doi: 10.1007/s40484-019-0187-4.
39. Yan Miao, Fu Liu, Tao Hou, and Yun Liu. Virtifier: A deep learning-based identifier for viral sequences from metagenomes. *Bioinformatics*, 38(5):1216–1222, 2022. doi: 10.1093/bioinformatics/btab845.
40. Yan Miao, Jilong Bian, Guanghui Dong, and Tianhong Dai. Detire: A hybrid deep learning model for identifying viral sequences from metagenomes. *Frontiers in Microbiology*, 14:1169791, 2023. doi: 10.3389/fmicb.2023.1169791.
41. Matthew P. Moore, Mirjam Laager, Paolo Ribeca, and Xavier Didelot. Kmeraperture: Retaining k-mer synteny for alignment-free extraction of core and accessory differences between bacterial genomes. *PLOS Genetics*, 20(4): e1011184, 2024. doi: 10.1371/journal.pgen.1011184.

42. Qiwen Yang, editor. *Practice and Progress of mNGS Report Interpretation*. Shanghai Science and Technology Press, 2026. Chinese-language edition; read-length context cited from p. 99.
43. Weichun Huang, Leping Li, Jason R. Myers, and Gabor T. Marth. Art: A next-generation sequencing read simulator. *Bioinformatics*, 28:593–594, 2012. doi: 10.1093/bioinformatics/btr708.
44. Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. doi: 10.2307/3001968.
45. Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
46. Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. Estimating mutual information. *Physical Review E*, 69 (6):066138, 2004. doi: 10.1103/PhysRevE.69.066138.
47. Brian C. Ross. Mutual information between discrete and continuous data sets. *PLOS ONE*, 9(2):e87357, 2014. doi: 10.1371/journal.pone.0087357.